Assignment – Mini Crowdfunding DApp

Objective

The goal of this assignment is to help you understand how real-world decentralized applications (DApps) work by building a **mini crowdfunding platform** on Ethereum. You will design, develop, and test a smart contract that allows users to create funding campaigns and accept contributions in a secure and transparent way.

This assignment focuses on **Solidity fundamentals, smart contract design, and real blockchain logic**, not just theory.

What You Have to Build

You need to create a **QuickStarter smart contract** that works like a simplified Kickstarter.

Core Features (Mandatory)

Your smart contract must support the following:

1. **Create a Campaign**
   - Campaign creator sets:
     - Campaign title
     - Funding goal (in ETH)
   - Reject campaigns with:
     - Zero funding goal
     - Past deadlines
2. **Contribute to a Campaign**
   - Any user can contribute ETH to an active campaign
   - Contribution amount must be greater than zero

Technical Requirements

- Language: **Solidity**
- Framework: **Hardhat**
- Use:
  - `struct` for campaign data
  - `mapping` to track contributions
  - `modifier` for access control
  - `require` statements for validations